昆山杜克大学 DUKE KUNSHAN UNIVERSITY

INFOSCI 301 · FINAL GROUP PROJECT · A0 LANDSCAPE

# A Source-Linked Visualization System for Comparing Recorded Color Patterns Across Places

For design/media students and creators, the prototype compares source-linked palettes across five places, revealing sample-specific—not culturally representative—color patterns.

Color Atlas · Tingyi Xiao—color algorithms · Jiaxin Wang—prototype and data · Andi Wan—visualization and interaction · Commit 754758f · Prof. Luyao Zhang · Autumn 2026

## 01 Question → user task → claim

| | |
|---|---|
| QUESTION | How can users compare recorded colors across places while tracing them to source photographs? |
| AUDIENCE | Design/media students, creators, and cultural educators. |
| TASK | Compare two samples and trace one similarity or difference to its source. |
| LOCAL EVIDENCE | Jinxi and Bacheng showed recurring muted tones, potentially shaped by weather, restoration, and sampling. |
| CURRENT CLAIM | The prototype compares source-linked palettes from 50 photographs across five places. |
| BOUNDARY | Results represent sampled photographs—not definitive local palettes or cultural similarity. |

SDG 11.4
Supports traceable communication of place-based visual materials; community benefit remains untested.

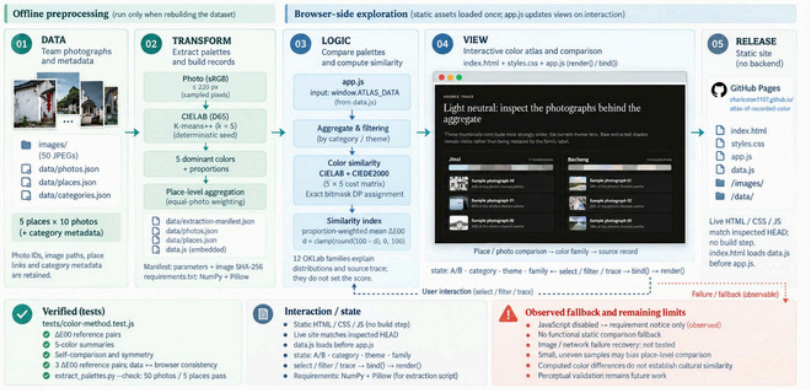
## FIG 1 Five idioms · one user task · five visible evidence panels

Same task: select two place or photo samples, compare recorded colors, and trace one difference to source evidence.

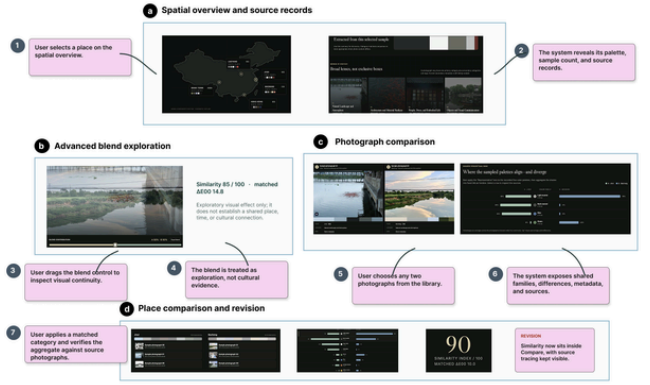



Data provenance: 50 team-authored photographs from Jinxi, Bacheng, Hong Kong, Genie Town, and Luoyang; original images and derived five-color palettes remain source-linked.
Access: public repository; team permission for course use; no project-wide reuse license. Missingness: small, uneven samples and incomplete capture metadata.

## FIG 2 Software architecture of the submitted version




## FIG 3 User-experience walkthrough → revision




## 02 Research → design → validation

**PRIMARY FRONTIER**
Human-centered spatiotemporal intelligence linking color, place, and source evidence.
RESEARCH → DESIGN → RESULT
Perceptual-color research informed K-means, CIELAB, and CIEDE2000 comparison with inspectable results.

**VALIDATION**
Domain: Local relevance remains untested.
Data/task: Source-linked but limited samples.
Idiom: Multiple views support comparison but may invite cultural overinterpretation.
Algorithm: Reproducible and symmetric; human agreement remains untested.

**NEXT CHECK**
Intended users complete a compare-and-trace task and explain the evidence boundary.

## 03 Full acknowledgements and credits

We thank Prof. Luyao Zhang for instruction and guidance in INFOSCI 301 and Michelle Clark, Jiajun Fu, Ruojin Wang, Enwei Fan, and Yiran Zheng for feedback on our methodology, cultural motivation, and community significance. Their comments led us to clarify the comparison, explain the algorithms, and better articulate how the project aligns with the UN Sustainable Development Goals. We thank our four community partners for fieldwork: Tencent's Shanghai Office, Shanghai Science and Technology Museum, Jinxi Intangible Cultural Heritage Museum, and China Ancient Brick and Tile Museum. Our observation of Suzhou water towns, including Jinxi during our September field trip and Bacheng, as well as photographs from our personal travels informed our decision to analyze the color palette that may give the artists and creators intuitive inspiration.We used ChatGPT (GPT-5.6) and Codex (GPT-5.6 Sol Medium), provided by OpenAI, to support code debugging and revision in index.html, interface wording, algorithm documentation, and selected text in the README.md, report, and poster. All outputs were reviewed, tested, and revised by the team. We also used original photographs taken by Andi Wan, Tingyi Xiao, and Jiaxin Wang with the photographers' permission in the artifact and project figures.

Andi Wan designed the network and spatial visualizations and refined the prototype's comparative data presentation; Jiaxin Wang developed the prototype framework, curated the image dataset, and implemented the core comparison interactions; Tingyi Xiao refined the color-extraction and comparison algorithms and implemented the advanced comparison effect.

## 04 Open the same submitted version

Working artifact/demo: https://charleston1107.github.io/atlas-of-recorded-color/; Repository + commit: https://charleston1107.github.io/atlas-of-recorded-color/; Poster: https://github.com/charleston1107/atlas-of-recorded-color/blob/main/ColorAtlas_INFOSCI301_Project_Poster.pdf; Paper: https://github.com/charleston1107/atlas-of-recorded-color/blob/main/ColorAtlas_INFOSCI301_Project_Report.pdf ; Demo video: https://github.com/charleston1107/atlas-of-recorded-color/blob/main/ColorAtlas_INFOSCI301_DemoVideo.mp4